

MAT1002 Final Examination

Monday, May 16, 2022

Time: 4:00 - 7:00 PM

Notes and Instructions

1. *This exam is closed-book. No book, note, dictionary, or calculator is allowed.*
2. *The total score of this examination is **110**.*
3. *There are **twelve** questions (with parts) in total.*
4. *The symbol $[N]$ at the beginning of a question indicates that the question is worth N points.*
5. *State your answers in exact form, e.g., write $\sqrt{2}$ instead of 1.414.*
6. *Show your intermediate steps **except Questions 1 and 2** — answers without intermediate steps will receive minimal (or even no) marks.*
7. *Onsite examinees should answer all questions in the answer book.*

MAT1002 Final Examination Questions

1. [6] True (T) or False (F)? No explanation is required.

- (i) Consider $f(x, y, z) = xy^2z^6 - \sin(e^{yz}) - \ln(x^2)$ defined on

$$D = \{(x, y, z) : 4 \leq x \leq 8, 3 \leq y \leq 4, -1 \leq z \leq 1\}.$$

Then there must exist (x_1, y_1, z_1) and (x_2, y_2, z_2) in D such that $f(x_1, y_1, z_1) \geq f(x, y, z) \geq f(x_2, y_2, z_2)$ for all $(x, y, z) \in D$.

- (ii) If a function $f(x, y)$ is differentiable at $(0, 0)$, then the partial derivatives f_x and f_y must both be continuous at $(0, 0)$.
- (iii) If the series $\sum_{n=1}^{\infty} u_n$ converges, then the series $\sum_{n=1}^{\infty} (u_{2n-1} - u_{2n})$ must also converge.

2. [27] Short questions. No explanation or intermediate steps are required.

- (i) Let $f(x, y, z) = x^2 + 2xy + yz$. Then $\operatorname{div}(\nabla f) = (\quad)$.
- (ii) Let $f(x, y, z) = \frac{1}{\sqrt{4-x^2-y^2-z^2}}$. Describe all the level surfaces of f .
- (iii) The temperature H is described by a differentiable function $H = f(x, y, z, t)$, where (x, y, z) is the position in the space and t is time. A space curve C is described by a smooth parametrization $x = x(t)$, $y = y(t)$, and $z = z(t)$, where t is time (same as above). What is the rate of change of temperature with respect to time along C at $t = 0$? Describe with an algebraic expression.
- (iv) Find the curvature of the curve $y = \sin x$ at the point $P(\pi/2, 1)$.
- (v) Find the principle unit normal for $y = -x^2$ at the point $P(1/2, -1/4)$.
- (vi) State the first three **nonzero** terms in the Maclaurin series of $\frac{x^2}{\sqrt{2+x}}$.
- (vii) Let $H = 5x^2 - 3xy + xyz$. At the point $P(3, 4, 5)$, in which direction does the value of H decrease the fastest? (You do not have to normalize your direction.)

- (viii) Let E be the solid that is outside the sphere $x^2 + y^2 + z^2 = z$ and inside the sphere $x^2 + y^2 + z^2 = 2z$. Write the following triple integral in the spherical coordinates (you do not need to compute the value):

$$\iiint_E z \, dV = (\quad).$$

- (ix) Let C be the circle that is the intersection of the sphere $x^2 + y^2 + z^2 = 1$ and the plane $x + y + z = 0$. Find the line integral $\int_C x^2 \, ds$.

3. [6+4=10] Consider the surface S given by $xz^2 - yz + \cos(xy) = 1$.

- (i) Find the tangent plane M and normal line ℓ to the surface S at the point $P(0, 0, 1)$.
- (ii) Show that the tangent line to the curve

$$\mathbf{r}(t) = (\ln t) \mathbf{i} + (t \ln t) \mathbf{j} + t \mathbf{k}$$

at $P(0, 0, 1)$ is lying on M .

4. [6] Let

$$f(x, y) = \begin{cases} \frac{x^2 y^2}{x^2 y^2 + (x - y)^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

Determine whether f is continuous at the origin. Justify.

5. [6] Find the global extrema of the function $f(x, y, z) = x + y + z$ subject to the constraints

$$x^2 + z^2 = 2 \quad \text{and} \quad x + y = 1.$$

6. [5+5] Evaluate the following integrals:

(i) $\int_0^1 \int_{3y}^3 e^{(x^2)} \, dx \, dy.$

(ii) $\int_0^1 \int_0^1 v(u + v^2)^4 \, du \, dv.$

7. [5] Use a **double integral** to find the area of the region inside the circle $(x - 1)^2 + y^2 = 1$ and outside the circle $x^2 + y^2 = 1$.

8. [5] Find the volume of the solid enclosed by the surface $(x^2 + y^2)^2 + z^2 = 1$.

9. [5+5] Suppose that the vector field

$$\mathbf{F} = (e^{kx} \ln y) \mathbf{i} + \left(\frac{e^{kx}}{y} + \sin z \right) \mathbf{j} + (my \cos z) \mathbf{k}$$

is conservative on $\{(x, y, z) : y > 0\}$, where k and m are two constants.

- (i) Find the values of k and m .
- (ii) Find one potential function of $\mathbf{F}$.

10. [5+3+5]

- (i) Suppose that C is a piecewise smooth, simple closed curve that is counterclockwise. Show that the area $A(R)$ of the region R enclosed by C is given by

$$A(R) = \oint_C x \, dy.$$

- (ii) Now consider the simple closed curve C in the xy -plane given by the polar equation $r = \sqrt{\sin \theta}$. State a parametrization of C .
- (iii) **Use the formula in part (i)** to find the area of the region enclosed by the curve C in part (ii).

11. [6] Let S be the cone $x = \sqrt{y^2 + z^2}$, $x \leq 2$. Given the vector field

$$\mathbf{F} = (\sin(x^3 y^2 z)) \mathbf{i} + (x^2 y) \mathbf{j} + (x^2 z^2) \mathbf{k},$$

find the flux done by the curl of $\mathbf{F}$ across S ,

$$\iint_S \text{curl}(\mathbf{F}) \cdot \mathbf{n} \, d\sigma,$$

with unit normals pointing to the positive x -direction.

12. [6] Let E be the solid that lies above the cone $z = \sqrt{\frac{x^2 + y^2}{3}}$ and below the sphere $x^2 + y^2 + z^2 = 1$, and let S be the boundary surface of E . Given the vector field

$$\mathbf{F} = (\cos(z^2)) \mathbf{i} + (z^3(y + x)) \mathbf{j} + (e^{x+y} x) \mathbf{k},$$

find the value of the outward flux

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, d\sigma.$$